

Data Analysis & Probability

ANSWER KEY



Mean, median, mode and range

- Find the mean of 6, 9, 12, 15, 18.
$$\text{Mean} = \frac{6 + 9 + 12 + 15 + 18}{5} = \frac{60}{5} = 12.$$
- Ages of 7 students: 13, 15, 13, 16, 14, 13, 17. Find the mean, median, and mode.
Ordered: 13, 13, 13, 14, 15, 16, 17. Mean = $\frac{101}{7} \approx 14.43$. Median (4th value) = 14. Mode = 13 (appears three times).
- Find the range and mode of the data set 8, 3, 8, 15, 8, 22, 3.
Range = $22 - 3 = 19$. Mode = 8 (appears three times).
- A data set has mean 18 and 6 values. Find the sum of the values.
Sum = $18 \times 6 = 108$.
- The data set 10, 12, 11, 13, 50 contains an outlier. Find the mean before and after removing 50.
Before: $\frac{10 + 12 + 11 + 13 + 50}{5} = \frac{96}{5} = 19.2$. After removing 50: $\frac{46}{4} = 11.5$.
- A data set of 5 values has mean 20. Four of the values are 15, 18, 22, 25. Find the fifth value.
Total sum = $20 \times 5 = 100$. Known sum = $15 + 18 + 22 + 25 = 80$. Fifth value = $100 - 80 = 20$.

Reading tables

- A survey of 50 people's favorite fruit: Apple 15, Banana 12, Orange 10, Mango 13. What fraction of people chose banana?
$$\frac{12}{50} = \frac{6}{25}.$$
- Using the same survey (Apple 15, Banana 12, Orange 10, Mango 13, out of 50), what percent chose either apple or mango?
 $15 + 13 = 28$ people. $\frac{28}{50} \times 100\% = 56\%.$
- A table shows quiz scores: score 5 (3 students), score 7 (5), score 8 (8), score 9 (4), score 10 (2). Find the mode score and the total number of students.
Mode = 8 (8 students, the highest frequency). Total students = $3 + 5 + 8 + 4 + 2 = 22$.
- Using the same quiz data, find the mean score to two decimal places.
Sum = $5(3) + 7(5) + 8(8) + 9(4) + 10(2) = 15 + 35 + 64 + 36 + 20 = 170$. Mean = $\frac{170}{22} \approx 7.73$.

Reading bar charts

- 11** The bar chart shows books sold by genre in one week. Find the total books sold and the modal (most common) genre.
Total = $45 + 30 + 20 + 15 = 110$ books. The modal genre is Fiction, with 45 books.
- 12** Using the same book-sales data, what percent of total books sold were mystery novels, to the nearest percent?
 $\frac{30}{110} \times 100\% \approx 27\%$.
- 13** The bar chart shows monthly rainfall in mm. Find the mean monthly rainfall.
Mean = $\frac{60 + 45 + 80 + 55}{4} = \frac{240}{4} = 60$ mm.
- 14** Using the same rainfall data (Jan 60, Feb 45, Mar 80, Apr 55 mm), find the percent decrease from March to April.
Decrease = $80 - 55 = 25$ mm. $\frac{25}{80} \times 100\% = 31.25\%$.
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Basic probability

- 15** A bag holds 4 red and 6 blue marbles. Find $P(\text{red})$.
 $P(\text{red}) = \frac{4}{10} = \frac{2}{5}$.
- 16** A fair six-sided die is rolled. Find $P(\text{a number greater than 4})$.
Favorable outcomes $\{5, 6\}$: $P = \frac{2}{6} = \frac{1}{3}$.
- 17** A card is drawn from a standard deck of 52 cards. Find $P(\text{a heart})$.
 $P = \frac{13}{52} = \frac{1}{4}$.
- 18** A spinner has 8 equal sections numbered 1 to 8. Find $P(\text{an even number})$.
Even outcomes $\{2, 4, 6, 8\}$: $P = \frac{4}{8} = \frac{1}{2}$.
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Combined and complementary events

- 19** A bag holds 3 red, 5 blue and 2 green marbles. Find $P(\text{not red})$.
 $P(\text{red}) = \frac{3}{10}$, so $P(\text{not red}) = 1 - \frac{3}{10} = \frac{7}{10}$.
- 20** Two fair coins are tossed. Find $P(\text{both land heads})$.
 $P = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.
- 21** A die is rolled twice. Find $P(\text{both rolls show a 6})$.
 $P = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$.
- 22** A bag holds 4 white and 6 black balls. One ball is drawn and not replaced, then a second is drawn. Find $P(\text{both white})$.
 $P = \frac{4}{10} \times \frac{3}{9} = \frac{12}{90} = \frac{2}{15}$.

Weighted averages

- 23** A class of 9 students has an average score of 80. A 10th student joins with a score of 100. Find the new average for all 10 students.
Old sum = $80 \times 9 = 720$. New sum = $720 + 100 = 820$. New average = $\frac{820}{10} = 82$.
- 24** Class A has 20 students with mean score 75; Class B has 30 students with mean score 85. Find the overall mean of all 50 students.
Total sum = $20(75) + 30(85) = 1,500 + 2,550 = 4,050$. Overall mean = $\frac{4,050}{50} = 81$.
- 25** A student's final grade is 40% homework average (score 90) and 60% exam average (score 70). Find the weighted final grade.
 $0.4 \times 90 + 0.6 \times 70 = 36 + 42 = 78$.
- 26** A survey combines two groups: 15 people with average age 24, and 25 people with average age 32. Find the overall average age of all 40 people.
Total sum = $15(24) + 25(32) = 360 + 800 = 1,160$. Overall mean = $\frac{1,160}{40} = 29$.

Comparing data sets

- 27** Data set A: 10, 12, 14, 16, 18. Data set B: 8, 14, 14, 14, 20. Both have the same mean. Which has the greater range?
Both have mean 14. Range of A = $18 - 10 = 8$. Range of B = $20 - 8 = 12$. Data set B has the greater range (more spread out).
- 28** Team X scored 22, 25, 28, 25 points in 4 games. Team Y scored 20, 20, 35, 25. Both teams have the same mean score. Which team's scores are more consistent (smaller range)?
Both have mean 25. Range of X = $28 - 22 = 6$. Range of Y = $35 - 20 = 15$. Team X is more consistent.
- 29** Class 1 has 18 students with mean score 68; Class 2 has 22 students with mean score 74. Find the combined mean of all 40 students, to one decimal place.
Total sum = $18(68) + 22(74) = 1,224 + 1,628 = 2,852$. Combined mean = $\frac{2,852}{40} = 71.3$.

Word problems

- 30** A teacher records the number of siblings for 10 students: 1, 2, 0, 1, 3, 2, 1, 0, 2, 1. Find the mean, median, and mode.
Sum = 13, mean = $\frac{13}{10} = 1.3$. Ordered: 0, 0, 1, 1, 1, 1, 2, 2, 2, 3; median = $\frac{1 + 1}{2} = 1$. Mode = 1 (appears four times).

- 31** A company's weekly sales (in thousands of riyals) over 5 weeks: 40, 55, 35, 60, 50. Find the mean weekly sales, and state how many weeks had sales above the mean.

Mean = $\frac{240}{5} = 48$ thousand riyals. Weeks above the mean: 55, 60, 50 — that is 3 weeks.

- 32** A jar contains 5 red, 3 blue and 2 yellow candies. A candy is picked and eaten, then another is picked. Find $P(\text{first red, then blue})$.

$$P = \frac{5}{10} \times \frac{3}{9} = \frac{15}{90} = \frac{1}{6}.$$

- 33** In a survey of 200 students, 120 prefer math and the rest prefer science. Find the probability, as a percent, that a randomly chosen student prefers science.

Science = $200 - 120 = 80$ students. $P = \frac{80}{200} = 0.4 = 40\%$.